

PRACTICAL NO. 4

AIM: Digital Signature

Implement digital signature algorithms such as RSA-based signatures, and verify the integrity and authenticity of digitally signed messages.

CODE:

```
import hashlib
```

```
def gcd(a, b):  
    while b:  
        a, b = b, a % b  
    return a
```

```
def mod_inverse(e, phi):  
    for d in range(1, phi):  
        if (e * d) % phi == 1:  
            return d  
    return None
```

```
def generate_keys():  
    # Two prime numbers  
    p = 61  
    q = 53  
  
    # Calculate n  
    n = p * q  
  
    # Euler's Totient  
    phi = (p - 1) * (q - 1)  
  
    # Public exponent  
    e = 17  
  
    # Private exponent  
    d = mod_inverse(e, phi)  
  
    return e, d, n
```

```
def hash_message(message):
```

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ROLL NO: T095

```
hash_value = hashlib.sha256(message.encode()).hexdigest()

hash_integer = int(hash_value, 16)

return hash_integer

def sign_message(message, d, n):
    hash_value = hash_message(message)

    hash_value = hash_value % n

    signature = pow(hash_value, d, n)

    return signature

def verify_signature(message, signature, e, n):
    hash_value = hash_message(message)
    hash_value = hash_value % n

    decrypted_hash = pow(signature, e, n)

    return decrypted_hash == hash_value

print("=====")
print("    RSA DIGITAL SIGNATURE")
print("=====")

e, d, n = generate_keys()

print("\nPublic Key :", (e, n))
print("\nPrivate Key :", (d, n))

message = input("\nEnter the message: ")

signature = sign_message(message, d, n)

print("\nDigital Signature:")
print(signature)

choice = input("\nDo you want to verify the signature? (yes/no): ")
```

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```
if choice.lower() == "yes":

    verify_message = input("\nEnter the message again for verification: ")

    try:
        entered_signature = int(
            input("Enter the digital signature: ")
        )

        if verify_signature(
            verify_message,
            entered_signature,
            e,
            n
        ):
            print("\n✓ DIGITAL SIGNATURE VALID")
            print("The message is authentic and unchanged.")

        else:
            print("\n✗ DIGITAL SIGNATURE INVALID")
            print("The message may have been modified.")

    except ValueError:
        print("\nInvalid signature. Please enter a number.")

else:
    print("\nVerification skipped.")

print("\n=====")
print("      PROGRAM END")
print("=====")
```

OUTPUT:

```

Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: D:/T095/T095 C.S. Prac 4.py =====
=====
RSA DIGITAL SIGNATURE
=====

Public Key : (17, 3233)
Private Key : (2753, 3233)

Enter the message: T095 AASHKA MISHRA

Digital Signature:
2368

Do you want to verify the signature? (yes/no): YES

Enter the message again for verification: T095 AASHKA
Enter the digital signature: 2368

X DIGITAL SIGNATURE INVALID
The message may have been modified.

=====
PROGRAM END
=====

>>>
===== RESTART: D:/T095/T095 C.S. Prac 4.py =====
=====
RSA DIGITAL SIGNATURE
=====

Public Key : (17, 3233)

X DIGITAL SIGNATURE INVALID
The message may have been modified.

=====
PROGRAM END
=====

>>>
===== RESTART: D:/T095/T095 C.S. Prac 4.py =====
=====
RSA DIGITAL SIGNATURE
=====

Public Key : (17, 3233)
Private Key : (2753, 3233)

Enter the message: T095 AASHKA MISHRA

Digital Signature:
2368

Do you want to verify the signature? (yes/no): YES

Enter the message again for verification: T095 AASHKA MISHRA
Enter the digital signature: 2368

✓ DIGITAL SIGNATURE VALID
The message is authentic and unchanged.

=====
PROGRAM END
=====

>>>
===== RESTART: D:/T095/T095 C.S. Prac 4.py =====

```

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```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help

Do you want to verify the signature? (yes/no): YES
Enter the message again for verification: T095 AASHKA MISHRA
Enter the digital signature: 2368

✓ DIGITAL SIGNATURE VALID
The message is authentic and unchanged.

=====
PROGRAM END
=====
>>>

===== RESTART: D:/T095/T095 C.S. Prac 4.py =====
=====
RSA DIGITAL SIGNATURE
=====

Public Key : (17, 3233)
Private Key : (2753, 3233)

Enter the message: T095 AASHKA MISHRA

Digital Signature:
2368

Do you want to verify the signature? (yes/no): NO
Verification skipped.

=====
PROGRAM END
=====
>>>
```

Ln: 74 Col: 0

28°C Partly sunny

Q Search

ENG IN

08:21 10-08-2026

NAME: AASHKA MISHRA
ROLL NO: T095

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GUI:

```
import tkinter as tk
from tkinter import messagebox
import hashlib
```

```
def gcd(a, b):
    while b:
        a, b = b, a % b
    return a
```

```
def mod_inverse(e, phi):
    for d in range(1, phi):
        if (e * d) % phi == 1:
            return d
    return None
```

```
def generate_keys():
```

```
    p = 61
```

```
    q = 53
```

```
    n = p * q
```

```
    phi = (p - 1) * (q - 1)
```

```
    e = 17
```

```
    d = mod_inverse(e, phi)
```

```
    return e, d, n
```

```
def get_hash(message):
```

```
    return hashlib.sha256(
        message.encode()
    ).hexdigest()
```

```
def sign_message(message, private_key, n):
```

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```
hash_value = int(
    get_hash(message),
    16
)

hash_value = hash_value % n

signature = pow(
    hash_value,
    private_key,
    n
)

return signature

def verify_signature(message, signature, public_key, n):

    original_hash = int(
        get_hash(message),
        16
    ) % n

    decrypted_signature = pow(
        signature,
        public_key,
        n
    )

    return original_hash == decrypted_signature

public_key, private_key, n = generate_keys()

BACKGROUND = "#F5F7FB"
WHITE = "#FFFFFF"

NAVY = "#172554"
BLUE = "#2563EB"
PURPLE = "#7C3AED"
LIGHT_PURPLE = "#F3E8FF"
```

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```
GREEN = "#16A34A"  
LIGHT_GREEN = "#DCFCE7"
```

```
RED = "#DC2626"  
LIGHT_RED = "#FEE2E2"
```

```
TEXT = "#172033"  
SECONDARY = "#64748B"  
BORDER = "#E2E8F0"
```

```
LIGHT_BLUE = "#EFF6FF"
```

```
def generate_signature():
```

```
    message = message_text.get(  
        "1.0",  
        tk.END  
    ).strip()
```

```
    if not message:
```

```
        messagebox.showwarning(  
            "Message Required",  
            "Please enter a message before generating the signature."  
        )
```

```
        return
```

```
    signature = sign_message(  
        message,  
        private_key,  
        n  
    )
```

```
    signature_text.delete(  
        "1.0",  
        tk.END  
    )
```


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```
signature_text.insert(
    tk.END,
    str(signature)
)

hash_text.delete(
    "1.0",
    tk.END
)

hash_text.insert(
    tk.END,
    get_hash(message)
)

status_label.config(
    text="✓ SIGNATURE GENERATED",
    fg=PURPLE
)

status_description.config(
    text="The message has been successfully signed using the RSA private key.",
    fg=SECONDARY
)

result_frame.config(
    highlightbackground=PURPLE
)

def verify():
    message = message_text.get(
        "1.0",
        tk.END
    ).strip()

    signature = signature_text.get(
        "1.0",
        tk.END
    ).strip()
```

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if not message:

```
    messagebox.showwarning(  
        "Message Required",  
        "Please enter a message."  
    )
```

```
    return
```

if not signature:

```
    messagebox.showwarning(  
        "Signature Required",  
        "Please generate or enter a digital signature."  
    )
```

```
    return
```

try:

```
    signature = int(signature)
```

except ValueError:

```
    messagebox.showerror(  
        "Invalid Signature",  
        "The digital signature must be a valid number."  
    )
```

```
    return
```

```
valid = verify_signature(  
    message,  
    signature,  
    public_key,  
    n  
)
```

```
hash_text.delete(
```

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```
"1.0",
tk.END
)

hash_text.insert(
    tk.END,
    get_hash(message)
)

if valid:

    status_label.config(
        text="✓ SIGNATURE VALID",
        fg=GREEN
    )

    status_description.config(
        text="Authenticity verified • Message integrity confirmed",
        fg=GREEN
    )

    result_frame.config(
        highlightbackground=GREEN
    )

    messagebox.showinfo(
        "Verification Successful",
        "DIGITAL SIGNATURE VALID\n\n"
        "✓ The message is authentic.\n"
        "✓ The message has not been modified."
    )

else:

    status_label.config(
        text="X SIGNATURE INVALID",
        fg=RED
    )

    status_description.config(
```

```
text="The message or digital signature may have been modified.",
fg=RED
)

result_frame.config(
    highlightbackground=RED
)

messagebox.showerror(
    "Verification Failed",
    "DIGITAL SIGNATURE INVALID\n\n"
    "⚠ The message may have been modified.\n"
    "⚠ Authenticity could not be confirmed."
)

def copy_signature():

    signature = signature_text.get(
        "1.0",
        tk.END
    ).strip()

    if not signature:

        messagebox.showwarning(
            "Nothing to Copy",
            "Generate a digital signature first."
        )

    return

root.clipboard_clear()

root.clipboard_append(
    signature
)

messagebox.showinfo(
    "Copied",
    "Digital signature copied successfully."
```

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```
)  
  
def clear_all():  
  
    message_text.delete(  
        "1.0",  
        tk.END  
    )  
  
    signature_text.delete(  
        "1.0",  
        tk.END  
    )  
  
    hash_text.delete(  
        "1.0",  
        tk.END  
    )  
  
    status_label.config(  
        text="● READY",  
        fg=SECONDARY  
    )  
  
    status_description.config(  
        text="Enter a message to begin the digital signing process.",  
        fg=SECONDARY  
    )  
  
    result_frame.config(  
        highlightbackground=BORDER  
    )  
  
root = tk.Tk()  
  
root.title(  
    "RSA Digital Signature | SecureTech"  
)  
  
root.geometry(  

```

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```
"1120x780"  
)  
  
root.resizable(  
    False,  
    False  
)  
  
root.configure(  
    bg=BACKGROUND  
)  
  
header = tk.Frame(  
    root,  
    bg=NAVY,  
    height=105  
)  
  
header.pack(  
    fill="x"  
)  
  
header.pack_propagate(False)  
  
icon = tk.Label(  
    header,  
    text="🔒",  
    font=("Segoe UI Emoji", 30),  
    bg=NAVY,  
    fg="white"  
)  
  
icon.pack(  
    side="left",  
    padx=(40, 15)  
)  
  
title_area = tk.Frame(  
    header,  
    bg=NAVY
```

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)

```
title_area.pack(  
    side="left"
```

)

```
title = tk.Label(  
    title_area,  
    text="RSA DIGITAL SIGNATURE",  
    font=("Segoe UI", 22, "bold"),  
    bg=NAVY,  
    fg="white"
```

)

```
title.pack(  
    anchor="w",  
    pady=(20, 0)
```

)

```
subtitle = tk.Label(  
    title_area,  
    text="Secure message authentication & integrity verification",  
    font=("Segoe UI", 10),  
    bg=NAVY,  
    fg="#CBD5E1"
```

)

```
subtitle.pack(  
    anchor="w"
```

)

```
system_status = tk.Label(  
    header,  
    text="● SYSTEM READY",  
    font=("Segoe UI", 10, "bold"),  
    bg="#1E3A8A",  
    fg="#BFDDBFE",  
    padx=16,
```

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```
pady=9
)

system_status.pack(
    side="right",
    padx=40
)

main = tk.Frame(
    root,
    bg=BACKGROUND
)

main.pack(
    fill="both",
    expand=True,
    padx=40,
    pady=25
)

key_card = tk.Frame(
    main,
    bg=WHITE,
    highlightthickness=1,
    highlightbackground=BORDER
)

key_card.pack(
    fill="x",
    pady=(0, 18)
)

key_heading = tk.Label(
    key_card,
    text="🔑 RSA KEY INFORMATION",
    font=("Segoe UI", 12, "bold"),
    bg=WHITE,
    fg=TEXT
)
```


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```
key_heading.grid(  
    row=0,  
    column=0,  
    columnspan=2,  
    sticky="w",  
    padx=22,  
    pady=(18, 12)  
)  
  
public_box = tk.Frame(  
    key_card,  
    bg=LIGHT_BLUE,  
    highlightthickness=1,  
    highlightbackground="#BFDDBFE"  
)  
  
public_box.grid(  
    row=1,  
    column=0,  
    sticky="ew",  
    padx=(22, 10),  
    pady=(0, 18)  
)  
  
public_title = tk.Label(  
    public_box,  
    text="PUBLIC KEY",  
    font=("Segoe UI", 9, "bold"),  
    bg=LIGHT_BLUE,  
    fg=BLUE  
)  
  
public_title.pack(  
    anchor="w",  
    padx=15,  
    pady=(12, 3)  
)
```

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```
public_value = tk.Label(  
    public_box,  
    text=f"(e = {public_key}, n = {n})",  
    font=("Consolas", 11),  
    bg=LIGHT_BLUE,  
    fg=TEXT  
)
```

```
public_value.pack(  
    anchor="w",  
    padx=15,  
    pady=(0, 12)  
)
```

```
private_box = tk.Frame(  
    key_card,  
    bg="#FFF7ED",  
    highlightthickness=1,  
    highlightbackground="#FED7AA"  
)
```

```
private_box.grid(  
    row=1,  
    column=1,  
    sticky="ew",  
    padx=(10, 22),  
    pady=(0, 18)  
)
```

```
private_title = tk.Label(  
    private_box,  
    text="PRIVATE KEY",  
    font=("Segoe UI", 9, "bold"),  
    bg="#FFF7ED",  
    fg="#EA580C"  
)
```

```
private_title.pack(  
    anchor="w",
```

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```
padx=15,  
pady=(12, 3)  
)  
  
private_value = tk.Label(  
    private_box,  
    text=f"(d = {private_key}, n = {n})",  
    font=("Consolas", 11),  
    bg="#FFF7ED",  
    fg=TEXT  
)  
  
private_value.pack(  
    anchor="w",  
    padx=15,  
    pady=(0, 12)  
)  
  
key_card.columnconfigure(  
    0,  
    weight=1  
)  
  
key_card.columnconfigure(  
    1,  
    weight=1  
)  
  
content = tk.Frame(  
    main,  
    bg=BACKGROUND  
)  
  
content.pack(  
    fill="both",  
    expand=True  
)
```

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```
message_card = tk.Frame(  
    content,  
    bg=WHITE,  
    highlightthickness=1,  
    highlightbackground=BORDER  
)
```

```
message_card.pack(  
    side="left",  
    fill="both",  
    expand=True,  
    padx=(0, 9)  
)
```

```
message_heading = tk.Label(  
    message_card,  
    text="✍ MESSAGE",  
    font=("Segoe UI", 13, "bold"),  
    bg=WHITE,  
    fg=TEXT  
)
```

```
message_heading.pack(  
    anchor="w",  
    padx=22,  
    pady=(20, 4)  
)
```

```
message_subtitle = tk.Label(  
    message_card,  
    text="Enter the message you want to digitally sign.",  
    font=("Segoe UI", 9),  
    bg=WHITE,  
    fg=SECONDARY  
)
```

```
message_subtitle.pack(  
    anchor="w",  
    padx=22  
)
```

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)

```
message_text = tk.Text(  
    message_card,  
    height=8,  
    font=("Segoe UI", 11),  
    bg="#F8FAFC",  
    fg=TEXT,  
    insertbackground=PURPLE,  
    relief="flat",  
    highlightthickness=1,  
    highlightbackground=BORDER,  
    highlightcolor=PURPLE,  
    padx=15,  
    pady=14,  
    wrap="word"
```

)

```
message_text.pack(  
    fill="both",  
    expand=True,  
    padx=22,  
    pady=14
```

)

```
hash_heading = tk.Label(  
    message_card,  
    text="SHA-256 HASH",  
    font=("Segoe UI", 9, "bold"),  
    bg=WHITE,  
    fg=SECONDARY
```

)

```
hash_heading.pack(  
    anchor="w",  
    padx=22
```

)

```
hash_text = tk.Text(  
    message_card,
```

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```
height=3,  
font=("Consolas", 9),  
bg="#F8FAFC",  
fg="#0369A1",  
relief="flat",  
highlightthickness=1,  
highlightbackground=BORDER,  
padx=12,  
pady=9,  
wrap="word"  
)
```

```
hash_text.pack(  
    fill="x",  
    padx=22,  
    pady=(6, 20)  
)
```

```
signature_card = tk.Frame(  
    content,  
    bg=WHITE,  
    highlightthickness=1,  
    highlightbackground=BORDER  
)
```

```
signature_card.pack(  
    side="right",  
    fill="both",  
    expand=True,  
    padx=(9, 0)  
)
```

```
signature_heading = tk.Label(  
    signature_card,  
    text="✍️ DIGITAL SIGNATURE",  
    font=("Segoe UI", 13, "bold"),  
    bg=WHITE,  
    fg=TEXT  
)
```

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```
signature_heading.pack(  
    anchor="w",  
    padx=22,  
    pady=(20, 4)  
)
```

```
signature_subtitle = tk.Label(  
    signature_card,  
    text="RSA signature generated from the message hash.",  
    font=("Segoe UI", 9),  
    bg=WHITE,  
    fg=SECONDARY  
)
```

```
signature_subtitle.pack(  
    anchor="w",  
    padx=22  
)
```

```
signature_text = tk.Text(  
    signature_card,  
    height=8,  
    font=("Consolas", 12),  
    bg="#FAF5FF",  
    fg=PURPLE,  
    insertbackground=PURPLE,  
    relief="flat",  
    highlightthickness=1,  
    highlightbackground="#DDD6FE",  
    highlightcolor=PURPLE,  
    padx=15,  
    pady=14,  
    wrap="word"  
)
```

```
signature_text.pack(  
    fill="both",  
    expand=True,  
    padx=22,
```

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```
pady=14
)

copy_button = tk.Button(
    signature_card,
    text="📋 COPY SIGNATURE",
    command=copy_signature,
    font=("Segoe UI", 9, "bold"),
    bg=LIGHT_PURPLE,
    fg=PURPLE,
    activebackground="#E9D5FF",
    activeforeground=PURPLE,
    relief="flat",
    cursor="hand2",
    padx=18,
    pady=8
)

copy_button.pack(
    anchor="e",
    padx=22,
    pady=(0, 20)
)

button_area = tk.Frame(
    main,
    bg=BACKGROUND
)

button_area.pack(
    fill="x",
    pady=20
)

generate_button = tk.Button(
    button_area,
    text="🔑 GENERATE SIGNATURE",
    command=generate_signature,
    font=("Segoe UI", 10, "bold"),
    bg=PURPLE,
```


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```
fg="white",
activebackground="#6D28D9",
activeforeground="white",
relief="flat",
cursor="hand2",
padx=25,
pady=11
)

generate_button.pack(
    side="left",
    padx=(0, 10)
)

verify_button = tk.Button(
    button_area,
    text="✓ VERIFY SIGNATURE",
    command=verify,
    font=("Segoe UI", 10, "bold"),
    bg=BLUE,
    fg="white",
    activebackground="#1D4ED8",
    activeforeground="white",
    relief="flat",
    cursor="hand2",
    padx=25,
    pady=11
)

verify_button.pack(
    side="left",
    padx=10
)

clear_button = tk.Button(
    button_area,
    text="🗑️ CLEAR",
    command=clear_all,
    font=("Segoe UI", 10, "bold"),
    bg=WHITE,
    fg=TEXT,
```

```
activebackground="#E2E8F0",
activeforeground=TEXT,
relief="flat",
highlightthickness=1,
highlightbackground=BORDER,
cursor="hand2",
padx=25,
pady=10
)

clear_button.pack(
    side="right"
)

result_frame = tk.Frame(
    main,
    bg=WHITE,
    highlightthickness=2,
    highlightbackground=BORDER
)

result_frame.pack(
    fill="x"
)

status_label = tk.Label(
    result_frame,
    text="● READY",
    font=("Segoe UI", 14, "bold"),
    bg=WHITE,
    fg=SECONDARY
)

status_label.pack(
    pady=(13, 2)
)

status_description = tk.Label(
    result_frame,
    text="Enter a message to begin the digital signing process.",
```

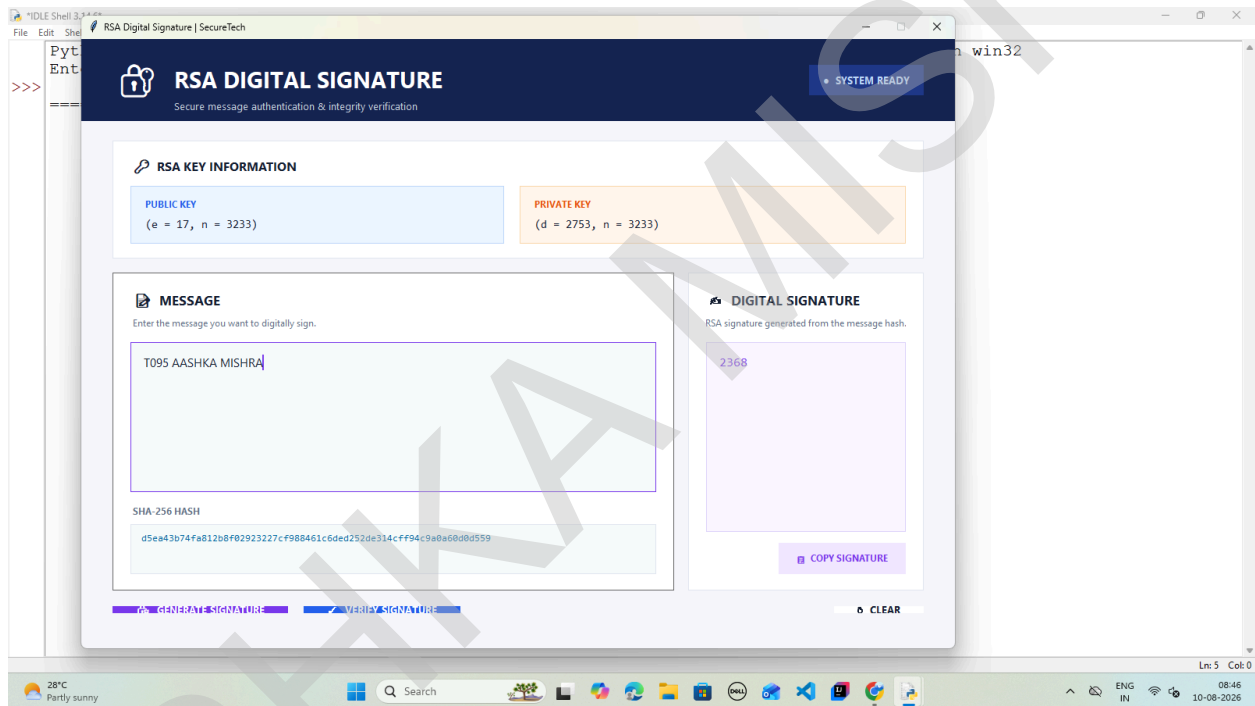
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```
font=("Segoe UI", 9),  
bg=WHITE,  
fg=SECONDARY  
)
```

```
status_description.pack(  
    pady=(0, 13)  
)
```

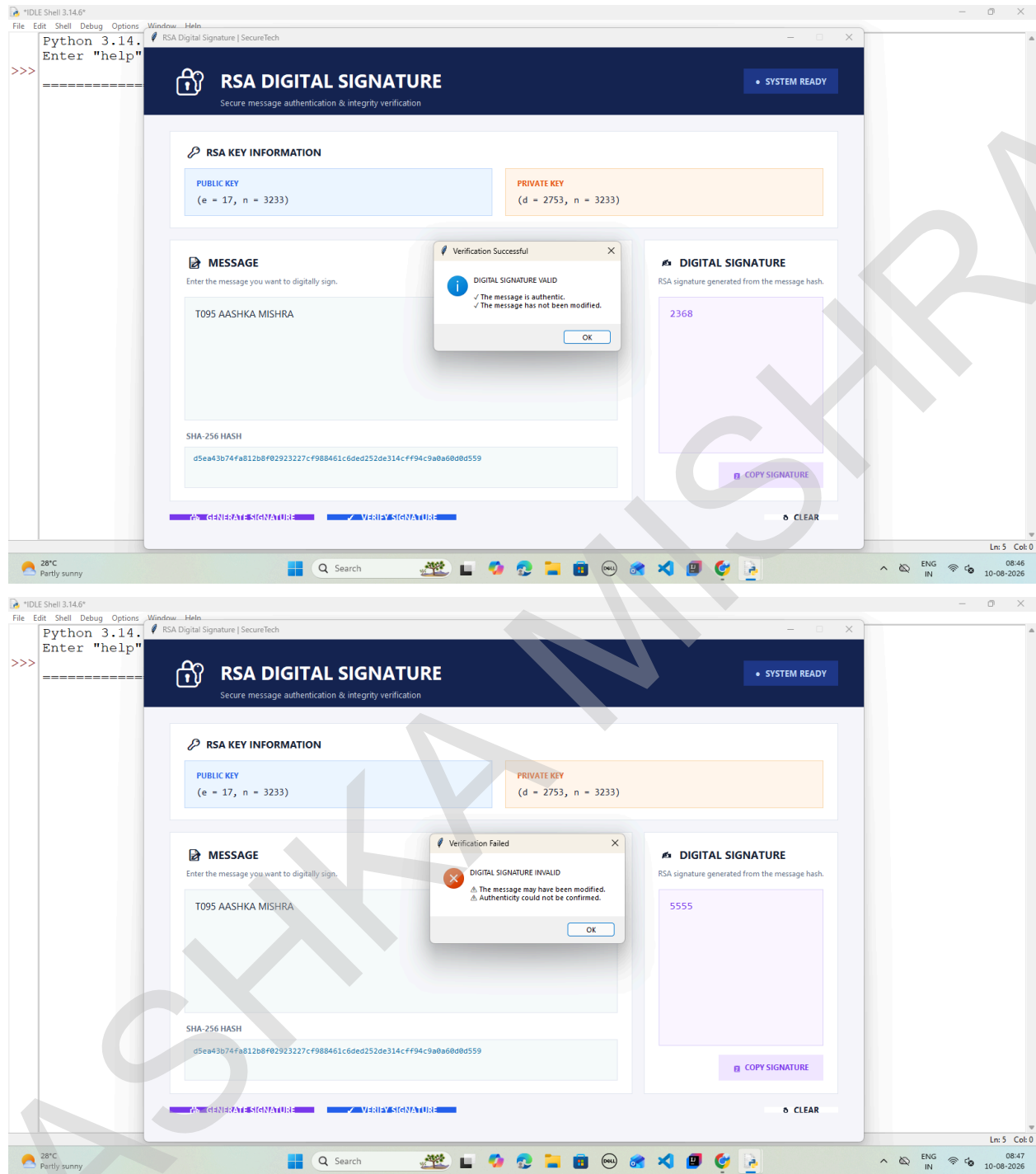
```
root.mainloop()
```

OUTPUT:



NAME: AASHKA MISHRA
ROLL NO: T095

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